

Pharmacology of NSAIDs

- Part 1A: Arachidonic acid, Cyclooxygenases, and the Biosynthesis of Prostanoids
- Part 1B: NSAIDs Class Effects

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I am available to groups and individuals for pharmacology help and discussions by appointment.

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After completing the preparation materials, students should be able to:

1. **Biochemistry / Physiology foundation of NSAID pharmacology:** Explain the synthesis of the biologically active prostanoids, PGE₂, PGI₂, PGF₂ α , PGD₂, and TXA₂, and their diverse receptor-mediated actions in various tissues.
2. **NSAIDs Class Pharmacokinetics:** Identify the NSAIDs class pharmacokinetic properties that are relevant for optimizing patients care.
3. **Mechanism of action:** Describe how inhibiting cyclooxygenase reduces the synthesis of biologically active prostanoids.
4. **Therapeutic uses:** Correlate the NSAID therapeutic uses with their relative affinities for COX-1- and COX-2 using specific examples.
5. **Adverse effects:** Link the NSAIDs' potentially serious adverse effects to the inhibition of COX-1 and COX-2 mediated prostanoid activity in the various tissues using specific examples.
6. **Drug interactions and Precautions:** Predict the potential drug interactions and precautions related to NSAID therapy, including in special populations (pregnancy and lactation, pediatric and geriatric patients, and patients with comorbidities).

Preparation Materials

Required

- Practice Questions (questions and case vignettes FOR learning)

Highly relevant optional materials:

- Video Lecture | Dr. Goldstein's Notes | Guided Reading Questions | ScholarRx Bricks
- Textbook resources with links are listed on the next slide and on the CPG

SUGGESTIONS:

- Use the resources that work best for you.
- You do not need to study all of them.
- Work through the GUIDED READING QUESTIONS with pen/pencil and paper.

Try answering them in your OWN WORDS first without looking at the readings, and then again after reviewing.

Links to resources

ScholarRx Brick: Neurology and Special Senses > Disorders of the Nervous System and Special Senses > Pain Mechanisms and Treatment > Diagnostic Approach to Disorders of the Nervous System and Special Senses

Title of Brick: Nonopioid Analgesics <https://exchange.scholarrx.com/brick/nonopioid-analgesics>

Title of Brick: Renal Effects of NSAIDs: <https://exchange.scholarrx.com/brick/renal-effects-of-nsaids>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 36: Nonsteroidal Anti-inflammatory Drugs, Disease Modifying Antirheumatic Drugs, Nonopioid Analgesics & Drugs Used in Gout > The section on Nonsteroidal Anti-inflammatory Drugs (NSAIDs)

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281752931>

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 18: The Eicosanoids: Prostaglandins, Thromboxanes, Leukotrienes, & Related Compounds

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281749744>

LWW Health Library, Premium Basic Sciences: Lippincott's Illustrated Reviews: Pharmacology, 8e, 2023; Chapter 40: Anti-inflammatory, Antipyretic, and Analgesic Agent

<https://nyit.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=253330080&bookid=3222>

Access Medicine Katzung's Pharmacology: Examination & Board Review, 14e, 2024; Chapter 36: NSAIDs, Acetaminophen & Drugs Used in Rheumatoid Arthritis & Gout

<https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?bookid=3382§ionid=281749744>⁵

NSAIDs Pharmacology Study Hints

First
The Big
Picture:
Class Effects



Then
How is each
drug the same
and different?
Drug-specific
Effects

1. Become familiar with the **physiological foundation** of the pharmacology.
2. Then, verbalize the **common thread** that stitches together all drugs in the group.
3. Then, learn the NSAIDs **CLASS** mechanisms / effects.
 - Pharmacokinetics
 - Mechanisms of action, mechanisms of adverse effects, mechanisms of drug interactions
 - Therapeutic uses
 - Drug interactions: drug-disease, drug-procedure, drug-drug
 - Contraindications – patient-specific effects related to comorbidities and drug interactions
4. **Then**, identify the starred (*) drugs of each subclass and how they represent the class effects. (Make the drug names your friends by saying them out loud.)
5. **Finally**, identify the drug-specific features of each drug: aspirin, salicylates, non-aspirin NSAIDs, acetaminophen

The **biochemistry and physiology** of arachidonic acid (AA) and conversion to the eicosanoids, the actions of the cyclooxygenases (COXs), the synthesis of prostanoids and their effects are **ESSENTIAL** for understanding the pharmacology of the NSAIDs.

Prostanoids: Subclass of potent autocrine / paracrine signaling molecules derived from AA

Prostaglandins: PGE₂, PGF₂ α , PGD₂

Other Prostanoids: PGI₂ (prostacyclin) and TXA₂ (thromboxane A₂)

- Produced in minute quantities by virtually all tissues
- Act locally on cells that secrete them or cells near the site of synthesis
- Brief action (~90 seconds)
- Bind various types of G protein-coupled receptors → specific functions, which depend on the tissue, enzyme pathways within the tissue, and receptors that are activated.

NSAIDs are chemically dissimilar compounds that inhibit COX-1 and/or COX-2.

- Decrease prostanoid synthesis
- Therapeutic uses for analgesic, anti-inflammatory, and antipyretic effects
- COX inhibition also results in potentially serious adverse effects due to prevention of constitutive and inducible effects of prostanoids.

Part 1A

The Biochemistry of Arachidonic Acid, The Biosynthesis of Prostanoids and Their Physiologic Effects

Definitions:

Eicosanoids: Very large family of signaling molecules synthesized by the enzymatic and nonenzymatic oxidation of arachidonic acid (AA) or other polyunsaturated fatty acids

AA is a 20-carbon polyunsaturated omega-6 fatty acid component of membrane phospholipids.

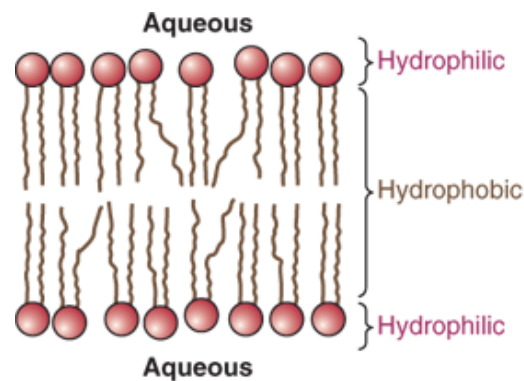
AA and other omega-6 fatty acids help maintain membrane fluidity, structural integrity, and cell signaling functions.

Prostanoids: A subclass of eicosanoids consisting of prostaglandins (PGE₂, PGF₂ α , PGD₂), prostacyclin (PGI₂), and thromboxane (TXA₂)

The BIG PICTURE: Prostanoid mediators derived from arachidonic acid and sites of drug action.

Arachidonic acid (AA) is present in intracellular membranes of the nuclear-endoplasmic reticulum complex.

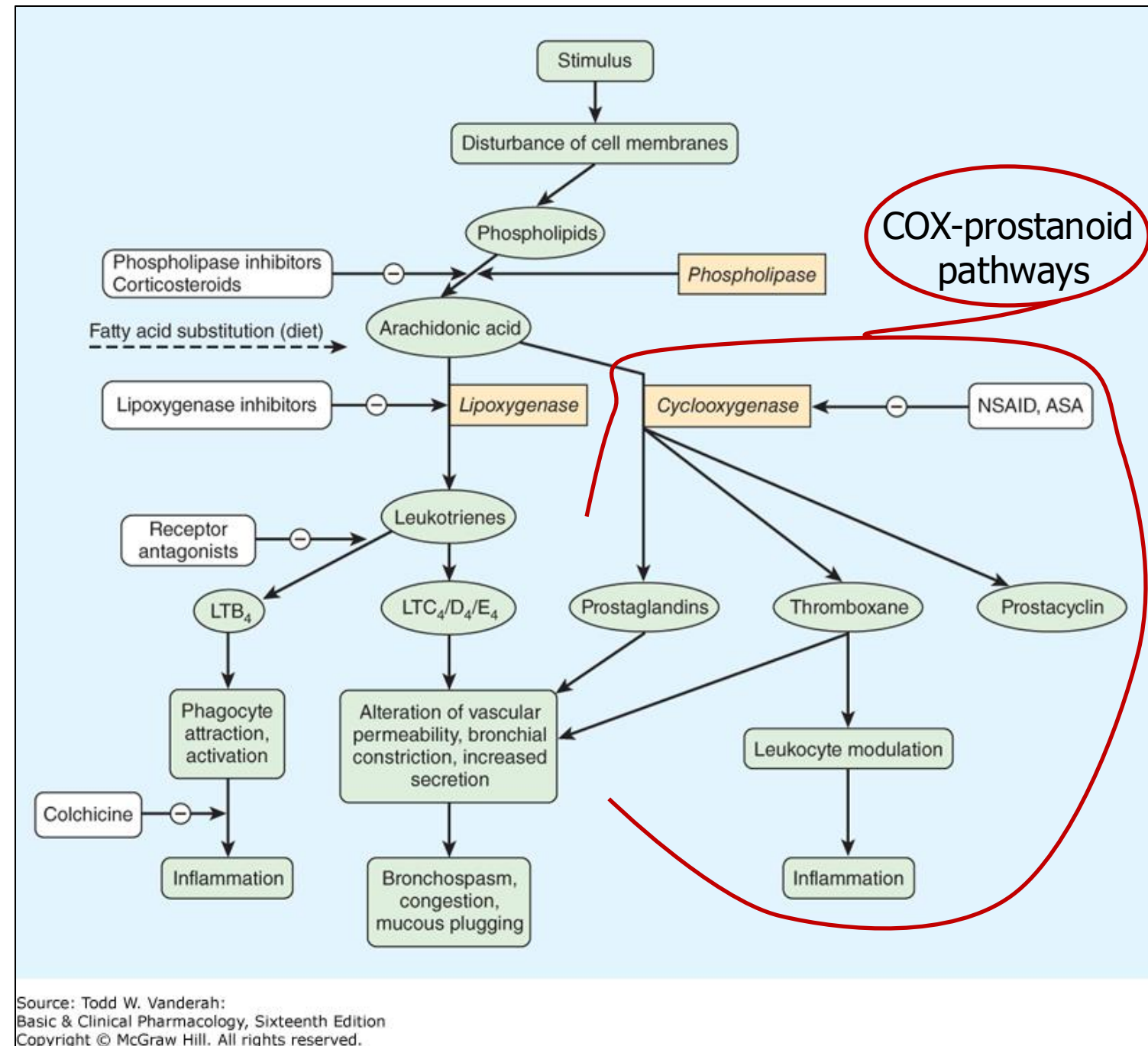
Chemical and physical stimuli activate phospholipase A₂ (PLA₂), which hydrolyzes membrane phospholipids, releasing AA.



Source: P.J. Kennelly, K.M. Botham, O.P. McGuinness, V.W. Rodwell, P.A. Weil: Harper's Illustrated Biochemistry, Thirty-second Edition Copyright © McGraw Hill. All rights reserved.

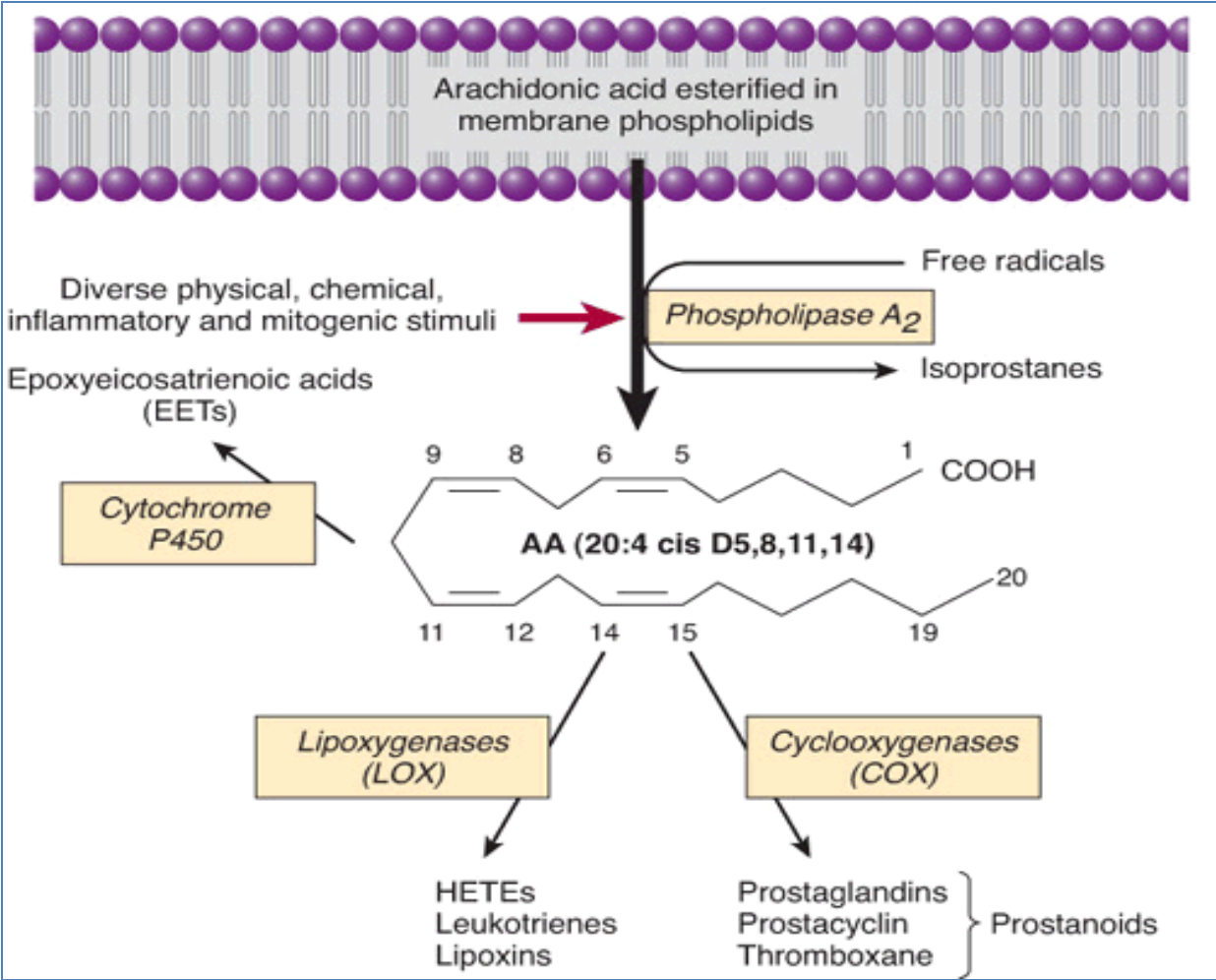
Diagram of a section of a bilayer membrane formed from phospholipids. The unsaturated fatty acid tails are kinked and lead to more spacing between the polar head groups, and hence to more room for movement. This in turn results in increased membrane fluidity.

Harper's Illustrated Biochemistry, 32e; 2023



ASA, acetylsalicylic acid (aspirin); LT, leukotriene; NSAID, nonsteroidal anti-inflammatory drug.

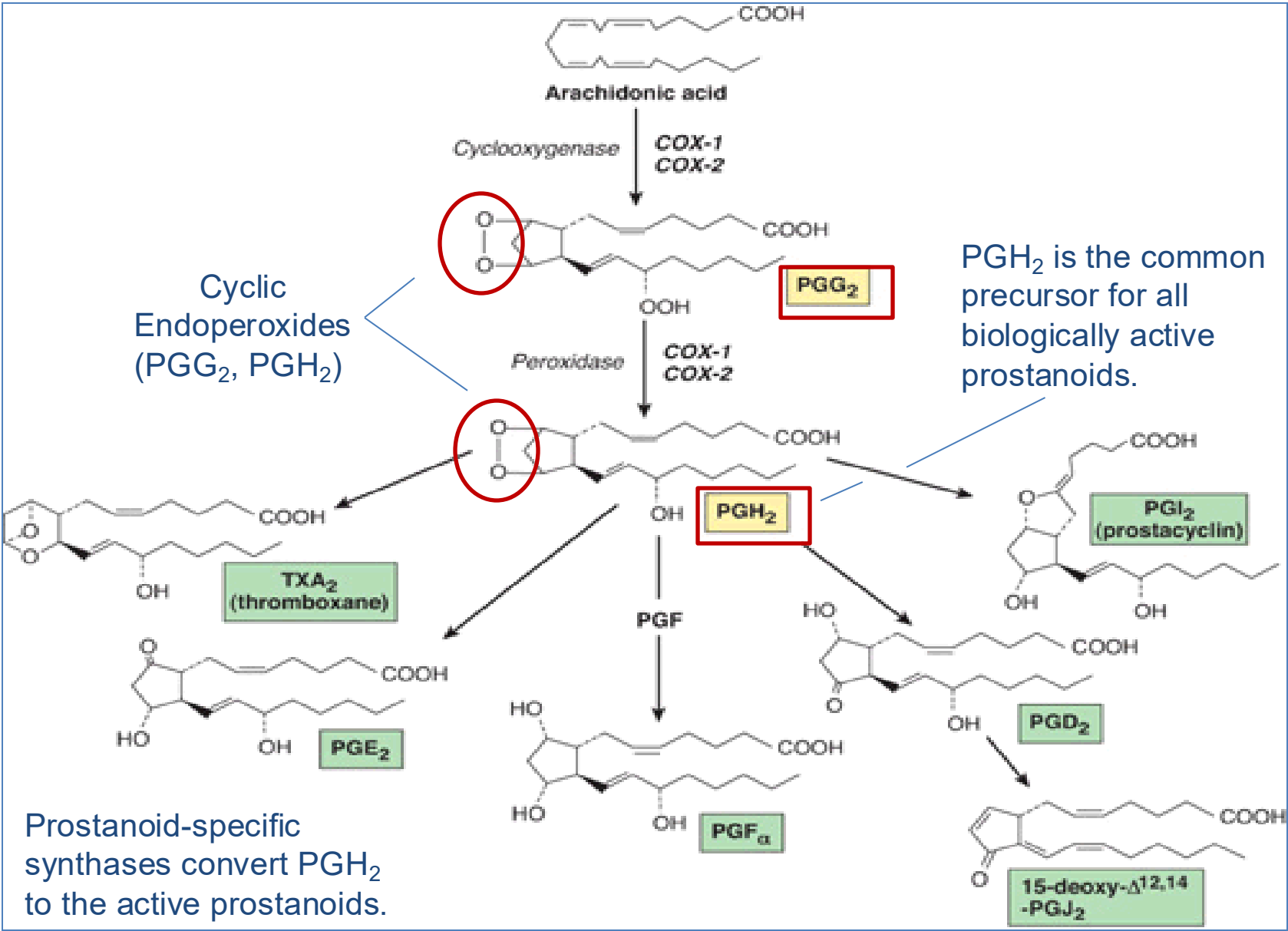
Arachidonic acid (AA) release and metabolism



Eicosanoids are oxygenation products of polyunsaturated long-chain fatty acids. Very large family; Highly potent compounds; Wide spectrum of biologic activity

Prostanoid biosynthesis

COX= prostaglandin endoperoxide synthase
a bifunctional enzyme (two catalytic sites)



Slide adapted by LG

AA substrate accesses the catalytic sites in COX via within the hydrophobic channel.

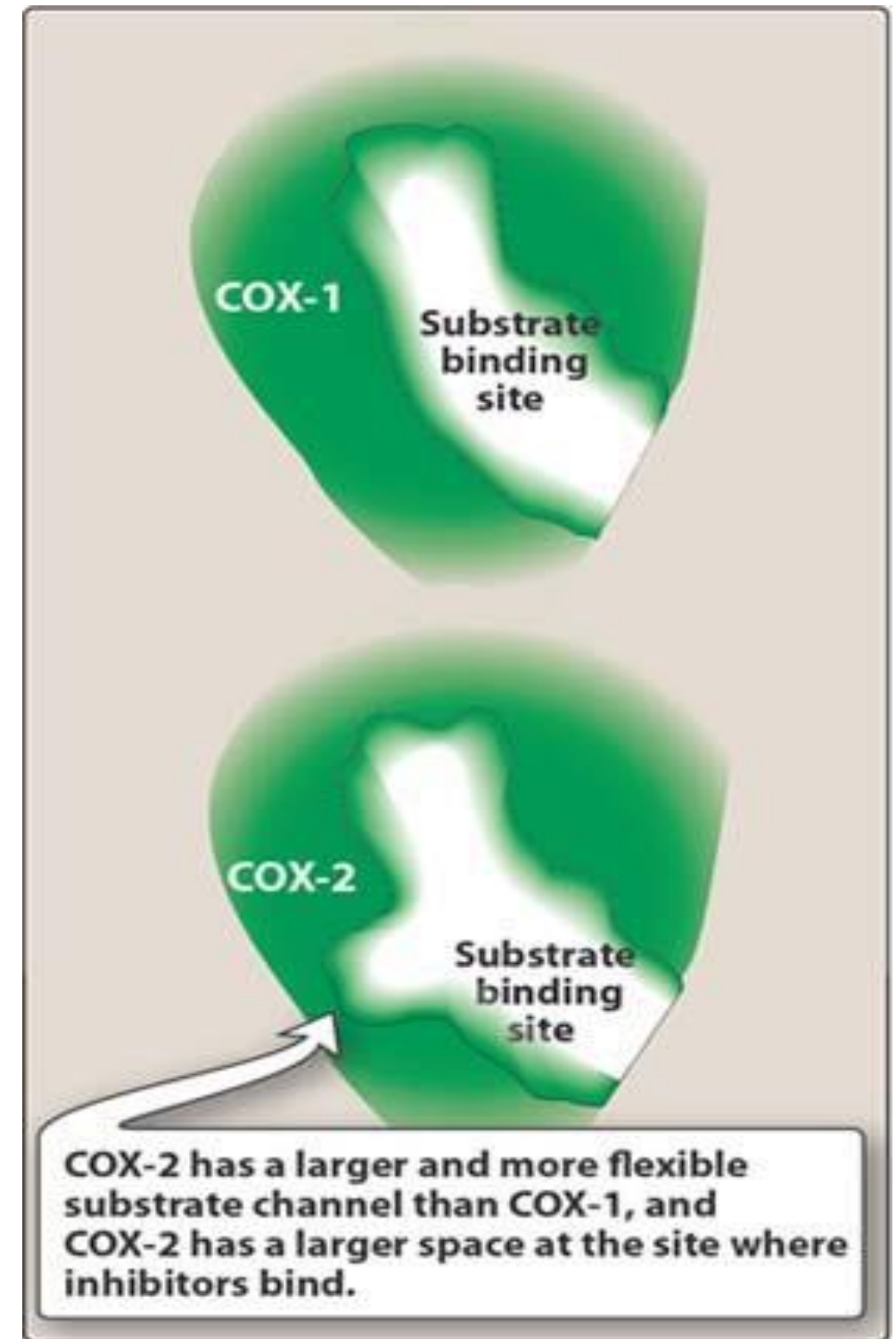
COX-1: constitutive; “housekeeping functions”

- Predominantly constitutively active = constantly active without an external or internal stimulus
- Maintains baseline prostaglandin synthesis for homeostatic (housekeeping) functions

COX-2: inducible; inflammation

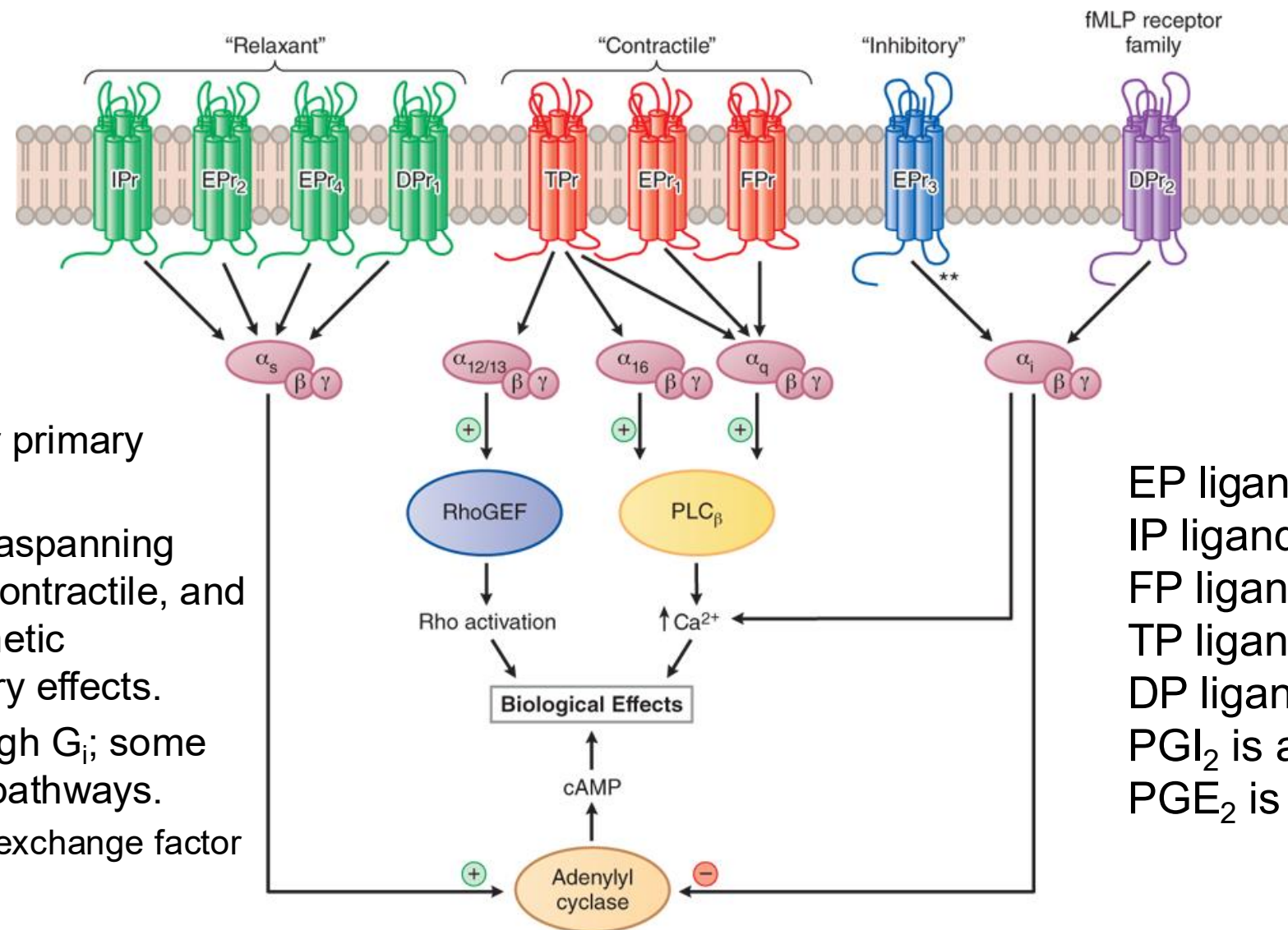
- Rapidly upregulated in response to inflammatory stimuli → inflammation, pain sensitization, fever response
 - Inflammatory stimuli include $\text{TNF}\alpha$, IL-1, growth factors, bacterial endotoxin

🔑 Both COX-1 and COX-2 contribute to the generation of **autoregulatory** and **homeostatic** prostanoids, and both can contribute to prostanoid formation during **inflammation**.



Prostanoid receptors are GPCRs.

TAKEAWAY: diverse, distinct receptors → diverse effects



Source: Laurence L. Brunton, Björn C. Knollmann:
Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e:
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EP ligand: PGE₂
 IP ligand: PGI₂
 FP ligand: PGF_{2α}
 TP ligand: TXA₂
 DP ligand: PGD₂
 PGI₂ is a secondary ligand of EP₁
 PGE₂ is a secondary ligand of IP

Legend:

Prostanoid receptors and their primary signaling pathways.

Prostanoid receptors are heptaspanning GPCRs. The terms relaxant, contractile, and inhibitory refer to the phylogenetic characterization of their primary effects.

All EP₃ isoforms couple through G_i; some can also activate G_s or G_{12/13} pathways.

RhoGEF: rho guanine nucleotide exchange factor

COX-1-derived Prostanoids → Homeostatic Functions of Prostanoids
predominantly constitutive actions

TAKEAWAY: NSAID use can disrupt the protective homeostatic functions of prostanoids leading to potentially serious adverse effects in at-risk individuals.
Several NSAIDs are available without a prescription – people self-treat.

GI tract cytoprotection PGE2 / PGI2	Stimulate mucus and bicarbonate secretion Maintain mucosal blood flow Reduce acid secretion Promote epithelial repair and turnover	Immune system PGD2	Maintains local tissue immune homeostasis Regulates mast cell function
Platelet function TXA2	Promote platelet aggregation Enhances platelet activation Causes local vasoconstriction Helps maintain homeostatic platelet plug	Vascular tone PGE2 / PGI2	Basal hemodynamics to support blood pressure: Contribute to basal vasodilation Counteract excessive vasoconstriction Support endothelial hemostasis
Renal autoregulation PGE2 / PGI2	Vasodilation of afferent arteriole Maintenance of glomerular filtration rate (GFR) Promote sodium and water excretion Renin release baseline regulation	Reproductive homeostasis	PGF2α uterine smooth muscle contraction (menstrual cycle endometrial changes) Initiation and maintenance of labor and supports postpartum involution
CNS PGE2 / PGD2	Microglial homeostasis Sleep regulation		

COX-2-derived prostanoid functions → Inflammation, pain, fever
predominantly inducible actions

Prostanoid	Inflammation	Pain	Fever
PGE2	Vasodilation Vascular permeability Immune modulation Tissue repair and regeneration	Major nociceptor sensitizer	Primary fever mediator Acts on hypothalamus to raise the body's temperature set point
PGI2	Vasodilation Vascular permeability Immune modulation	Nociceptor sensitizer	_____
PGD2	Derived from mast-cell COX-2 Allergic inflammation	Mild sensitization	_____
TXA2	Vasoconstriction Platelet activation	_____	_____

Check your knowledge

What membrane-derived fatty acid is the precursor of the prostanoids?

Where in the cell is the precursor located?

What is the process that converts the precursor into the five biologically active prostanoids?

Name the five prostanoids.

Check your knowledge

What determines whether certain prostanoids are constitutively expressed or inducible? (and define the terms)

How do the prostanoids exert their different effects in various body tissues?

The end of Part 1A: the biochemistry of arachidonic acid,
the biosynthesis of prostanoids and their physiologic effects

Pharmacology of NSAIDs, Part 1B

NSAIDs Class Effects

- Pharmacokinetic properties
- Mechanisms
- Therapeutic use
- Toxicities
- Pregnancy and Lactation
- Drug interactions

NSAIDs

SALICYLATES

- *Aspirin and active salicylic acid metabolite
 - Diflunisal
 - other non-aspirin salicylates

ACETIC ACID DERIVATIVES

- *Indomethacin
- *Ketorolac
- *Diclofenac
 - Sulindac
 - Etodolac
 - Nabumetone
 - Tolmetin

***ACETAMINOPHEN**

(not an NSAID)

PROPIONIC ACID DERIVATIVES

- *Ibuprofen
- *Naproxen
 - Fenoprofen
 - Ketoprofen
 - Flurbiprofen
 - Oxaprozin

FENAMATES

- Mefenamic acid
- Meclofenamate

ENOLIC ACID DERIVATIVES

- *Meloxicam
- Piroxicam

COX-2 SELECTIVE (COXIB)

- *Celecoxib

Pharmacokinetics: Class Properties

Chemically heterogeneous; weak acids (pka typically 3-5)
(except nabumetone, a nonacidic prodrug)

A	Rapid, complete, not affected by food onset of analgesia 30-60 minutes; $T_{\max}$ 1-4 hrs
D	~99% albumin bound; cross blood-brain barrier, accumulate at sites of inflammation and synovial fluid

rapid
pain relief

accumulate
at site of action

Analgesic effects are related to CNS actions: reduced central sensitization, modulation of descending pain pathways, and possibly interaction with other systems. Accumulation in synovial fluid is directly related to NSAIDs' effectiveness in arthritis.

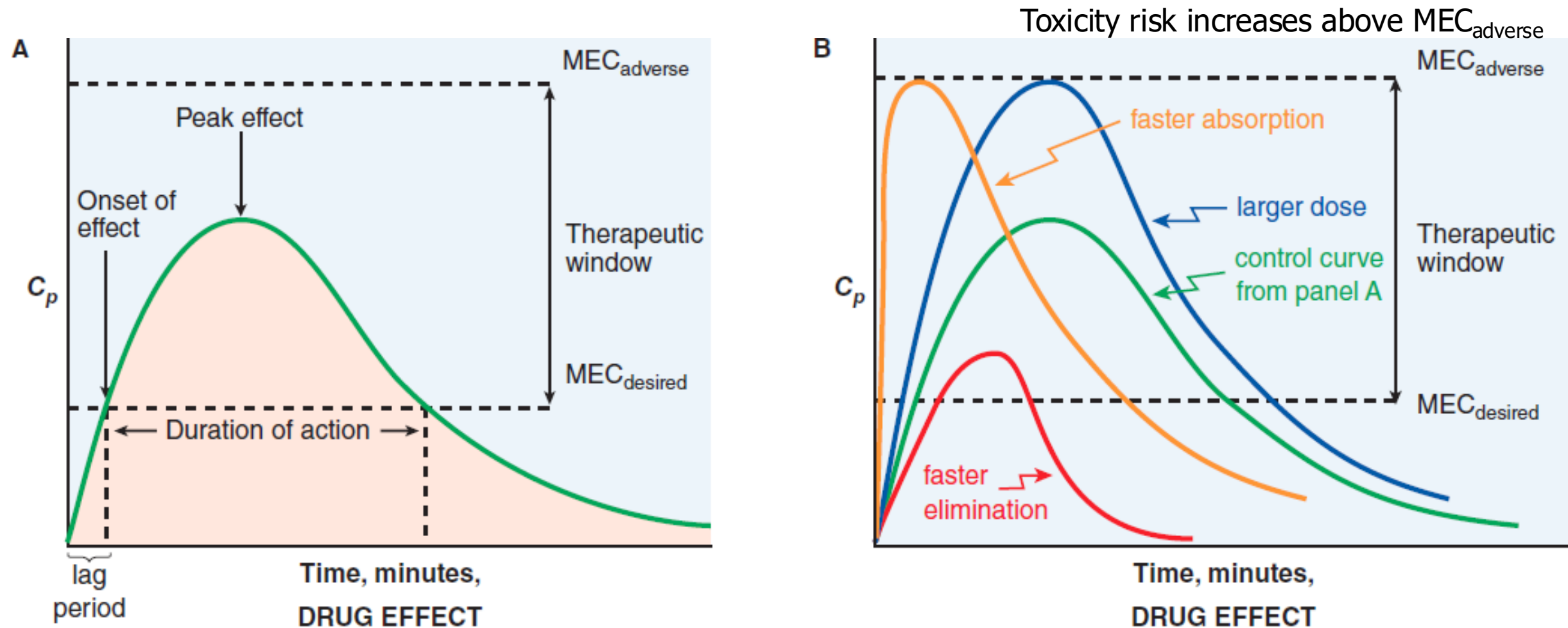
M	Hepatic, minor substrates of CYPs, esp CYP3A, CYP2C families; glucuronidation; enterohepatic circulation
E	glomerular filtration / tubular secretion of metabolites (small amount unchanged drug)
t_{1/2}	< 6 hours to > 10 hours

Caution impaired
organ fx

short, moderate, or long
durations of action

Pharmacokinetics Refresher

NSAIDs have dose-/concentration-dependent effects.



Source: Laurence L. Brunton, Randa Hilal-Dandan, Björn C. Knollmann: Goodman & Gilman's: The Pharmacological Basis of Therapeutics, Thirteenth Edition: Copyright © McGraw-Hill Education. All rights reserved.

MEC: Minimum effective concentration | C_p : plasma concentration | Shaded area, panel A: Area under the curve, AUC

MOA: NSAIDS inhibit COX → ↓ biosynthesis of prostanoid precursors

- **Aspirin**

- **Irreversibly** acetylates a serine residue within the hydrophobic channel of COX-1 and COX-2 → creates steric hindrance that blocks AA entry to the catalytic channel.
- Aspirin is the **only** NSAID that is an irreversible inhibitor.

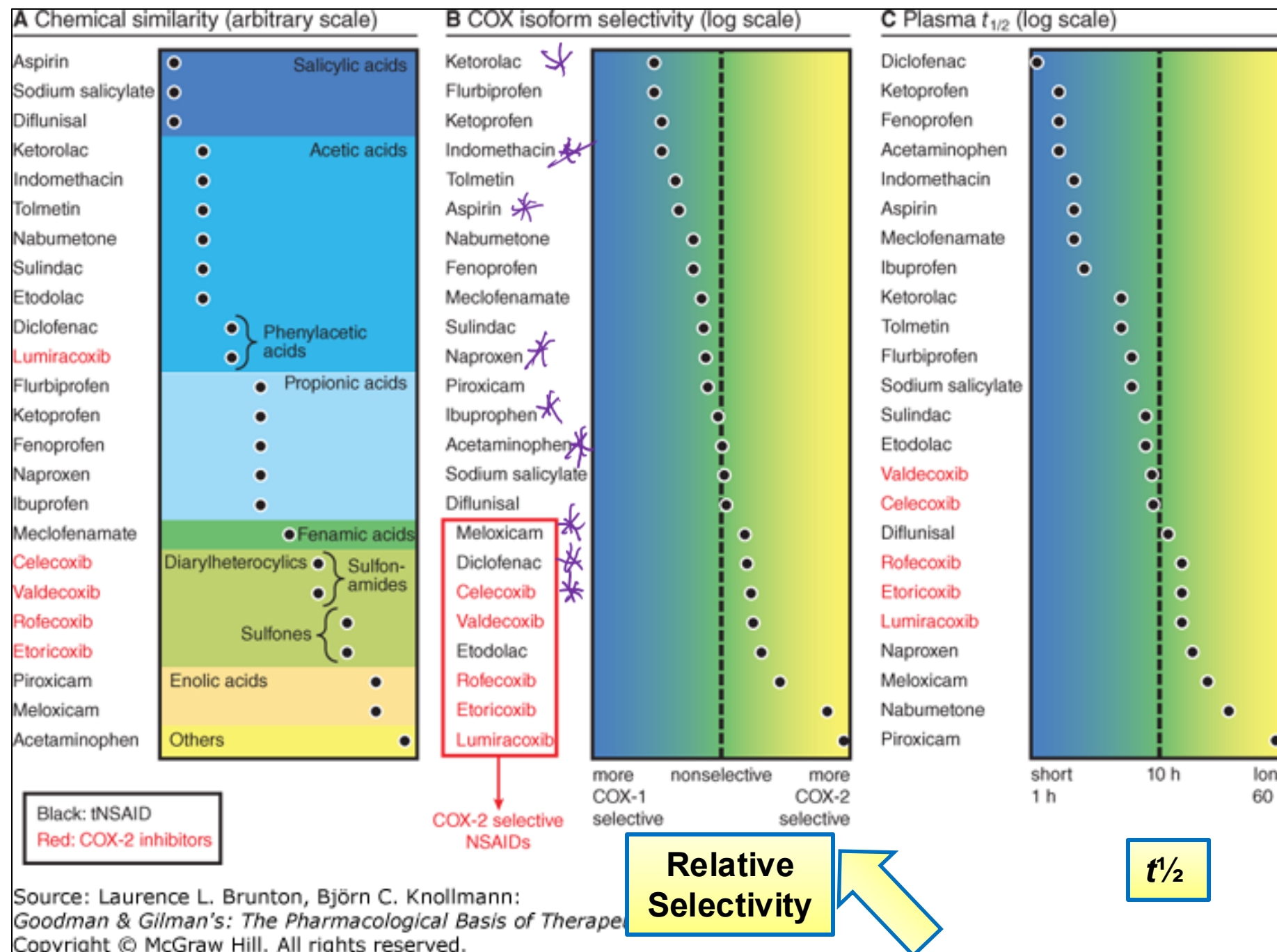
- **Non-aspirin NSAIDs**

- **Reversible inhibitors** of COX-1 and COX-2 → block AA access to catalytic site
 - Some nonspecific NSAIDs (nsNSAIDs) exhibit greater affinity for COX-2 over COX-1 (diclofenac, meloxicam, and etodolac)

- **“Coxibs”**

- **Selective** inhibitors of COX-2 – bulky molecule creates steric hindrance that blocks AA entry
 - Coxibs fit the larger binding pocket of COX-2.
 - Bulky structure does not fit narrower COX-1 channel.

Additional anti-inflammatory mechanisms of NSAIDs described in the literature:
COX-independent interference with neutrophil activation and chemotaxis → inhibits release of mediators of inflammation.



Pharmacodynamics (panel B)

NSAIDs' effects are related to their relative selectivity for COX-1/COX-2

1. nsNSAIDs:
nonselective (ns) inhibitors of both COX-1 and COX-2
2. Coxibs:
selective COX-2 inhibitors
3. Diclofenac, Meloxicam, Etodolac are more selective for COX-2 than for COX-1.

Classification of NSAIDs by chemical similarity (A), COX isoform selectivity (B), and plasma $t_{1/2}$ (C). The COX selectivity chart is plotted from data published in Warner T., et al. Nonsteroid drug selectivities for cyclooxygenase-1 rather than cyclooxygenase-2 are associated with human gastrointestinal toxicity: a full in vitro analysis. Proc Natl Acad Sci USA, 1999, 96:7563–7568; and FitzGerald GA, Patrono C. The coxibs, selective inhibitors of cyclooxygenase-2. N Engl J Med, 2001, 345:433–442.

NSAIDs: Mechanism → Effect Relationship

Therapeutic effects:

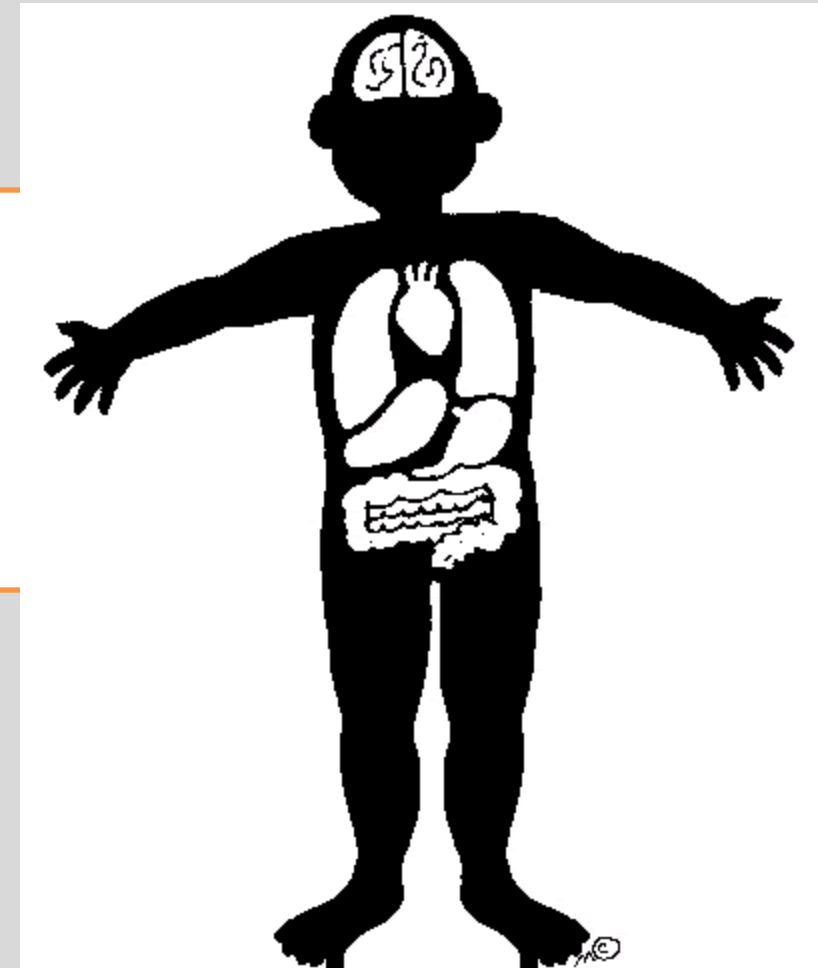
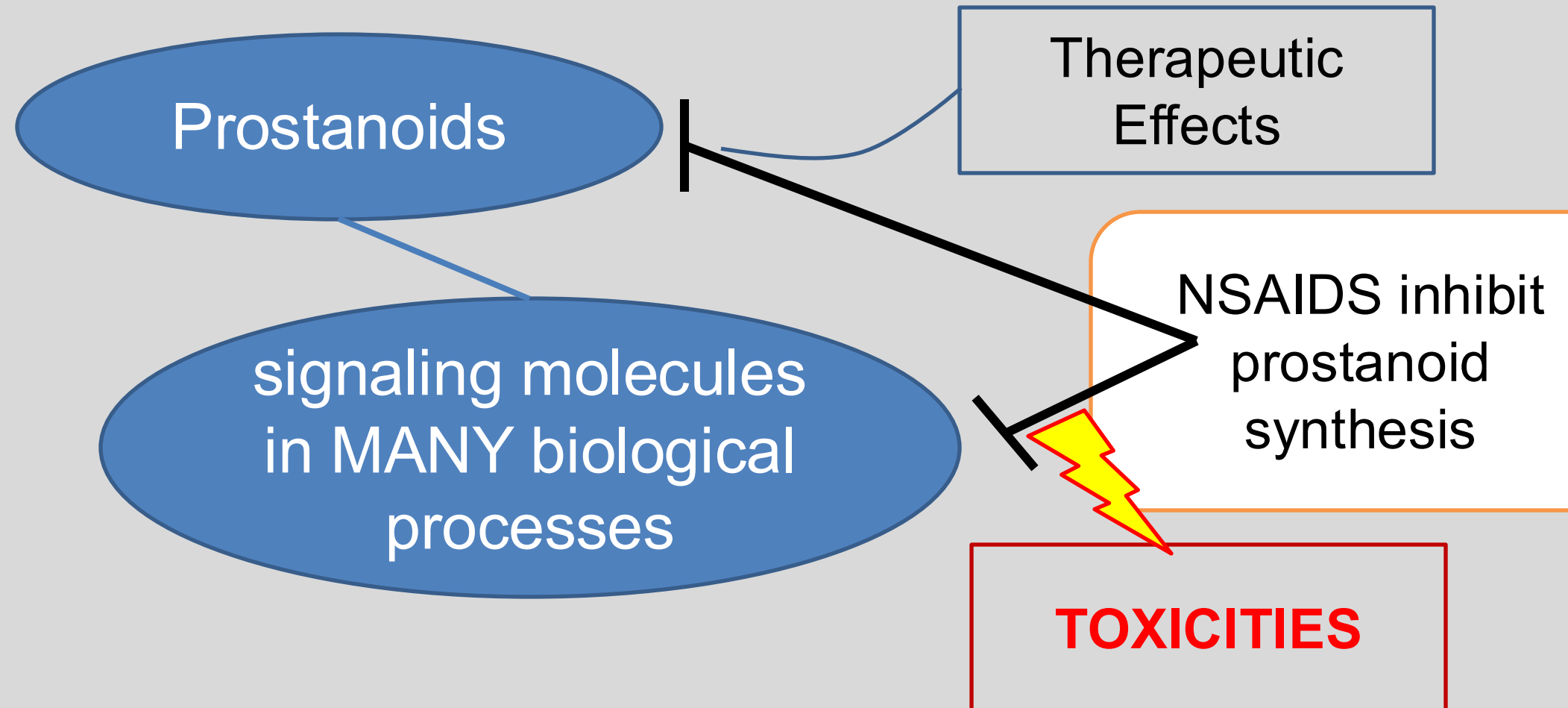
- **COX-2 inhibition** → analgesic, antipyretic, and anti-inflammatory actions

Adverse effects:

- **COX-1 inhibition** → prevents synthesis of GI cytoprotective PGE2 and PGI2
- **COX-1 inhibition in platelets** → TXA2 synthesis inhibition → reduced platelet aggregation (desirable effect only with low dose aspirin)
- **COX-2 inhibition in endothelial cells** → PGI2 synthesis inhibition → inhibits counterbalancing platelet disaggregation → increased risk of bleeding AND increased risk of thrombosis
- **COX-1 and COX-2 inhibition** → renal PGE2 and PGI2 synthesis inhibition → decreases renal blood flow and sodium / water excretion (and other effects)

What is the “common thread” that ties all these drugs together?

Please lift your eyes away from the PPT slide and try to answer this question.



All NSAIDs treat pain, fever, and inflammation.

All NSAIDs can cause potentially serious adverse effects in vulnerable patients.

Therapeutic Uses



Pain

NSAIDs

Acetaminophen



Fever

NSAIDs

Acetaminophen

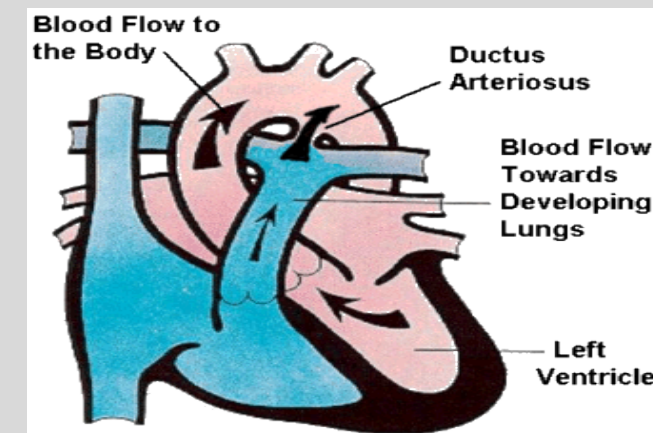


Inflammation

NSAIDs but
not acetaminophen

Class Therapeutic Effects and Uses

- Analgesia
- Antipyretic
- Anti-inflammatory
- Antiplatelet
 - Only aspirin is used for CV benefit.
- Ophthalmic inflammation
 - Topical ophthalmic drops
- Closure of the ductus arteriosus
 - Premature neonates
- Niacin tolerability (PGD2)
- PG upregulation disorders
 - Systemic mastocytosis
 - Bartter and Gittelman syndromes



🔑 All NSAIDs are ~ equally efficacious
Differentiate by cost & AEs

Upper figure: <http://www.desimd.com/health-education/eye-disorders/allergic-conjunctivitis-seasonal-allergy>

Lower figure: http://www.marvistavet.com/html/patent_ductus_arteriosus.html

Check your knowledge

How do NSAIDs inhibit the synthesis of biologically active prostanoids?

How does the NSAIDs' mechanism of action relate to their therapeutic and adverse effects?

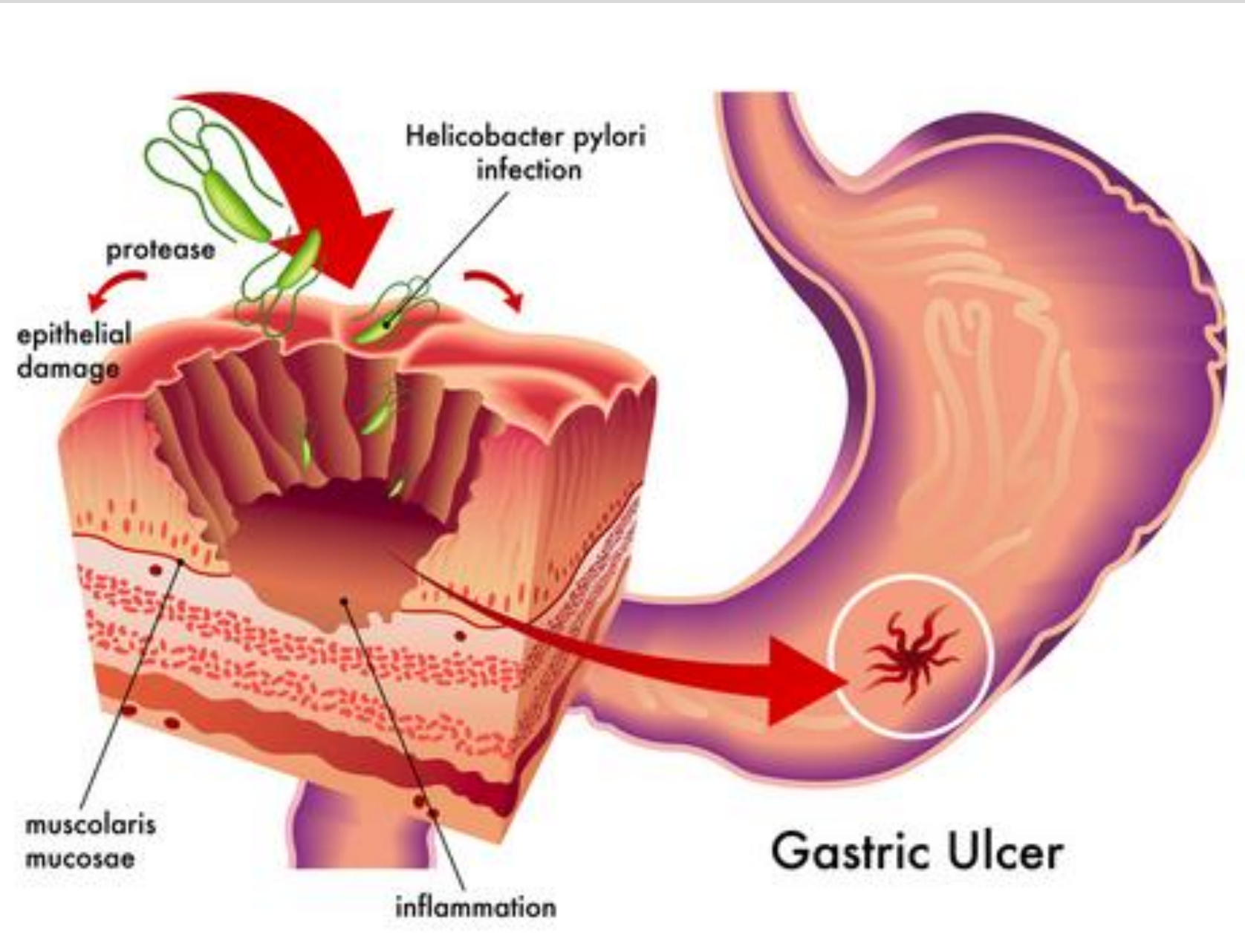
Why is it helpful to recognize the affinities of the individual NSAIDs for COX-1 and COX-2?

NSAIDS TOXICITIES

Serious adverse effects resulting from prostaglandin inhibition

- Gastrointestinal
- Cardiovascular
- Renal

Gastrointestinal adverse effects: Result from inhibition of COX-1-derived cytoprotective actions of PGE2 and PGI2.



NSAID use inhibits protective and repair processes → can cause:

- Gastric irritation
- Peptic ulcer
- GI bleeding
- Perforation

Adverse effects can occur throughout the entire GI tract.

Risk factors:

1. *Helicobacter pylori* infection
2. mucosal injury, including inflammatory bowel disease
3. alcohol consumption
4. concurrent anticoagulant
5. concurrent glucocorticoid use

NSAID + GC → reduce mucosal repair
→ major ↑risk

Cardiovascular adverse effects result from inhibition of platelet COX-1-derived TXA₂ and endothelial COX-2-derived PGI₂.

Platelet Thromboxane A₂
(TXA₂)

- NSAIDs inhibit platelet COX-1-mediated TXA₂ synthesis →
- Inhibits local platelet aggregation and vasoconstriction →
 - Inhibits formation of the platelet plug →
 - **Increases the risk of bleeding**

Endothelial Prostacyclin
(PGI₂)

- NSAIDs inhibit endothelial COX-2-mediated PGI₂ →
- Inhibits counterbalancing local vasodilation and platelet disaggregation
 - Endothelial COX-2 is more sensitive to NSAID inhibition than platelet COX-1 →
 - **Shifts the balance to platelet plug formation → prothrombotic effects**

Activated Platelets

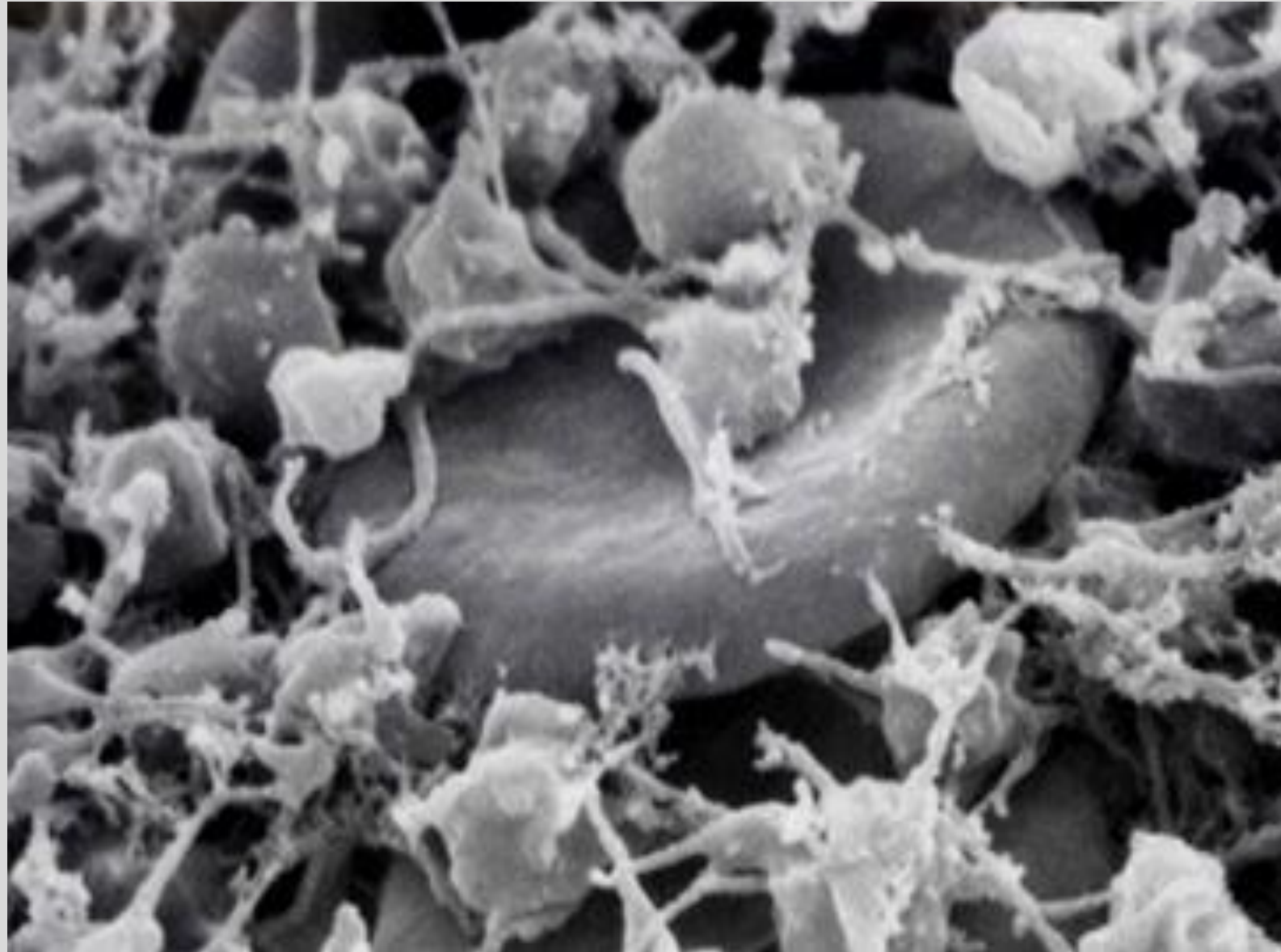


Figure: activated platelets

http://biomed.brown.edu/Courses/BI108/BI108_2005_Groups/10/pictures/web/platelets1.jpg

Aspirin (protective)	non-aspirin nsNSAIDs (increased risk of thrombosis)	Selective COX-2 inhibitors (increased risk of thrombosis)
Low-dose aspirin inhibits platelet COX-1 mainly in the portal system. Much of the dose is metabolized on the first pass through the liver.	nsNSAIDs achieve therapeutic blood levels in the systemic circulation.	COX-2 selective NSAIDs block endothelial COX-2 in the systemic circulation but do not block platelet COX-
This action shifts the balance towards antiplatelet effects when aspirin is taken daily.	- Block both COX-1 in platelets & COX-2 in endothelial cells. - COX-2 more sensitive to block → shift balance towards platelet thrombus formation, particularly under inflammatory/shear stress conditions	This shifts the balance towards platelet thrombus formation.

Actions of prostanoids on vascular tissues that are inhibited by NSAIDs

		Low-dose Aspirin	non-aspirin nsNSAIDs	COX-2 selective
CV tissue	CV action of prostaglandins	Irreversible Inhibition	Reversible Inhibition	
Platelets (COX-1)	Induces platelet aggregation	✓	✓	
	Induces vasoconstriction by TXA ₂	✓	✓	
Endothelial cells (COX-2)	Inhibits platelet aggregation		✓	✓
	Induces vasodilation by PGE ₂ , PGI ₂		✓	✓

Low-dose aspirin is cardioprotective.

Non-aspirin NSAIDs and COX-2 inhibitors INCREASE risk of adverse CV events.

Clinical aspects of NSAIDs-related cardiovascular risk: atherothrombosis, myocardial infarction, and stroke (except with aspirin)

Chronic or repeated NSAID use increases risk of thrombosis.

BP elevation and heart failure exacerbation contribute to CV risk.

1. Higher dose / longer duration → ↑ risk.
2. COX-2 selective and non-aspirin nsNSAIDs are both associated with increased CV risk.

PRACTICE POINTS

1. Low dose / Short duration
2. Reserve COX-2 selective drugs for high GI risk
3. Consider alternate treatment for high-risk patients

Risk factors of coronary heart disease and stroke include tobacco use, alcohol use, high blood pressure (hypertension), high cholesterol, obesity, physical inactivity, and unhealthy diets.

US Boxed Warning: All NSAIDs (except aspirin) are contraindicated during perioperative cardiopulmonary bypass graft surgery → ↑ thrombotic and CV risk.

Renal Effects: First – physiology

Nephron filters the blood; regulates fluid, electrolytes, waste

Afferent arteriole controls how much blood enters the glomerulus.

Efferent arteriole controls how much pressure builds up inside the glomerulus.

Together they fine-tune RBF and GFR.

Autoregulatory effects of prostanoids

Cortical COX-2-derived PGE₂ and PGI₂ → arteriolar vasodilation → **maintains local RBF and GFR**

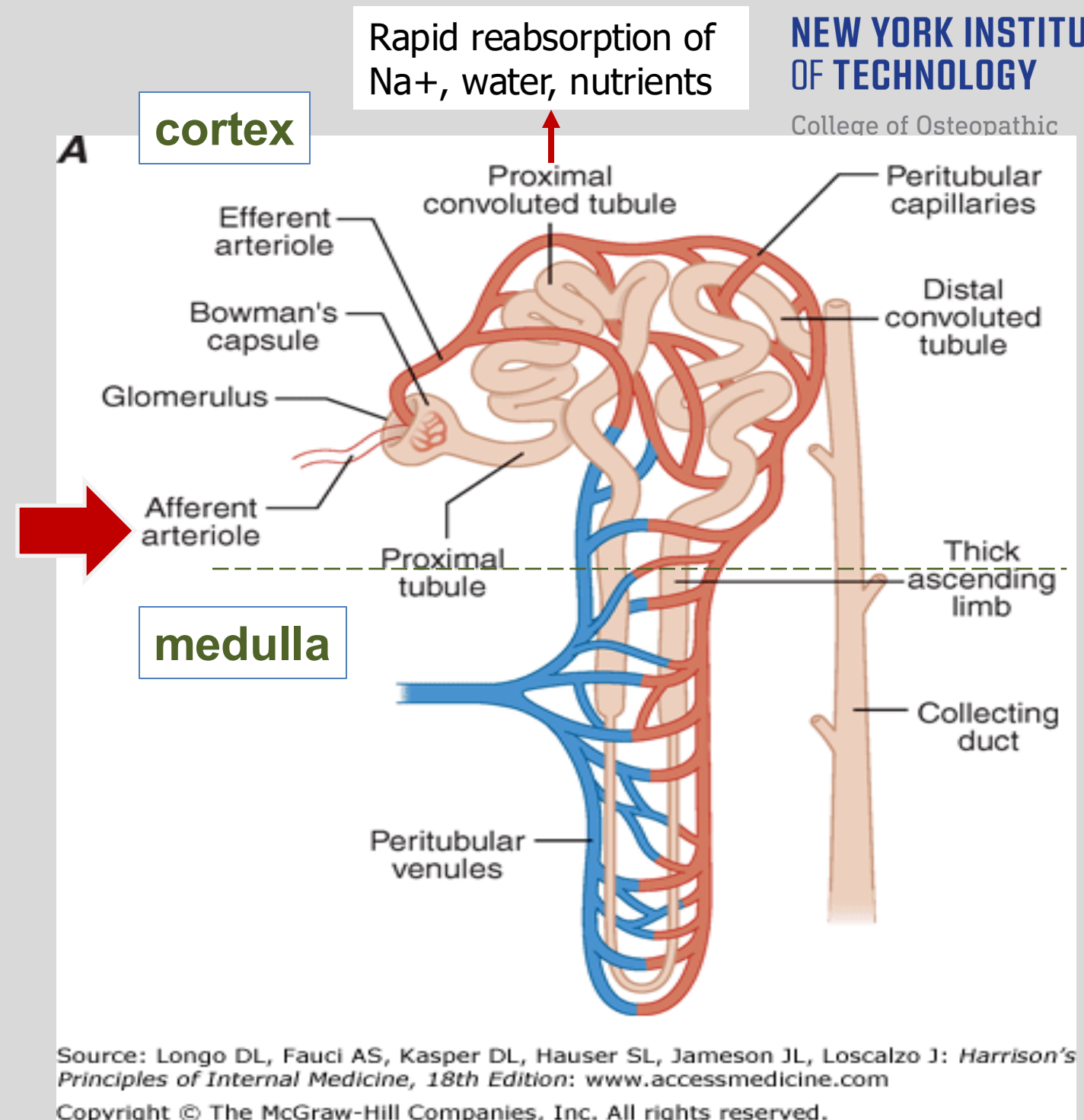
Medullary COX-2-derived PGE₂ and PGI₂ → **↑ medullary blood flow and ↓ tubular Na⁺ reabsorption**

COX-1-derived PGE₂ → **promote salt and water excretion in collecting ducts**

PGE₂-stimulated **renin release in volume-contracted states**

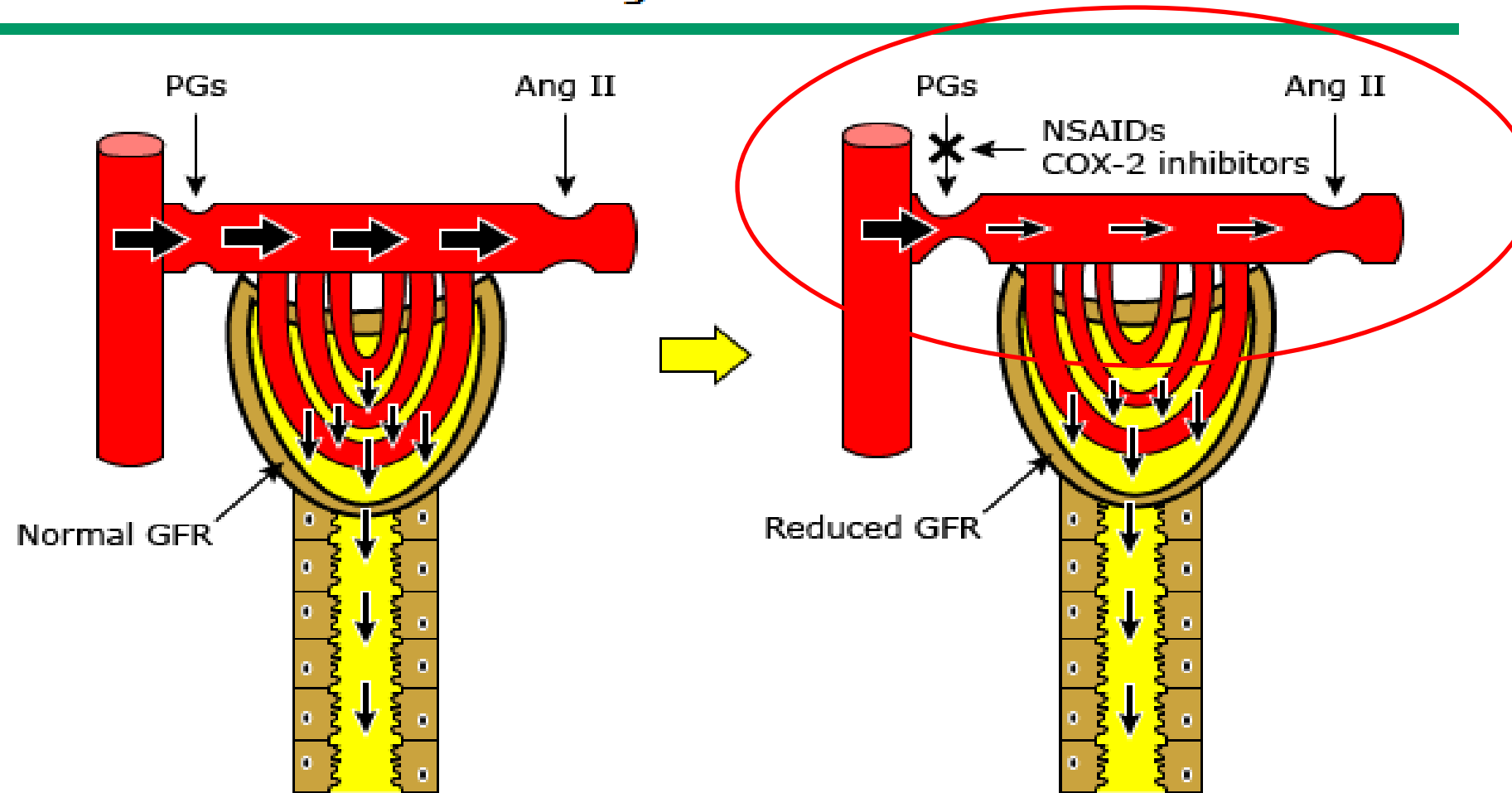
RBF: renal blood flow

GFR: glomerular filtration rate



NSAIDs block the autoregulatory effects of PGs.

NSAIDs block the vasodilatory effect of prostaglandins on the afferent arteriole and reduce glomerular filtration rate



Certain conditions that promote renal vasoconstriction and renal hypoperfusion require prostaglandins to vasodilate the afferent arteriole and blunt vasoconstriction to maintain a normal GFR (left). However in the presence of NSAIDs, which reduced prostaglandin production, afferent arteriole vasoconstriction dominates and reduces GFR (right).

GFR: glomerular filtration rate; PGs: prostaglandins; Ang II: angiotensin II; NSAIDs: nonsteroidal anti-inflammatory drugs; COX: cyclooxygenase.

Courtesy of Randy Luciano, MD, PhD, and Mark A Perazella, MD, FACP.

Now: Link the Prostanoids – NSAIDs Mechanism – Renal Adverse Effects

Inhibition of prostanoid synthesis →

1. Afferent arteriole vasoconstriction
 - Reduces renal perfusion
 - Worsens GFR / renal function
 - Increases proteinuria
2. Causes salt and water retention
 - Hypertensive complications (new onset or worsened control)
 - Increased risk of edema
 - Hyperkalemia ($\uparrow$ plasma K^+)

Risk factors:

Prostanoid formation becomes crucial in patients with marginally functioning kidneys and volume-contracted states.

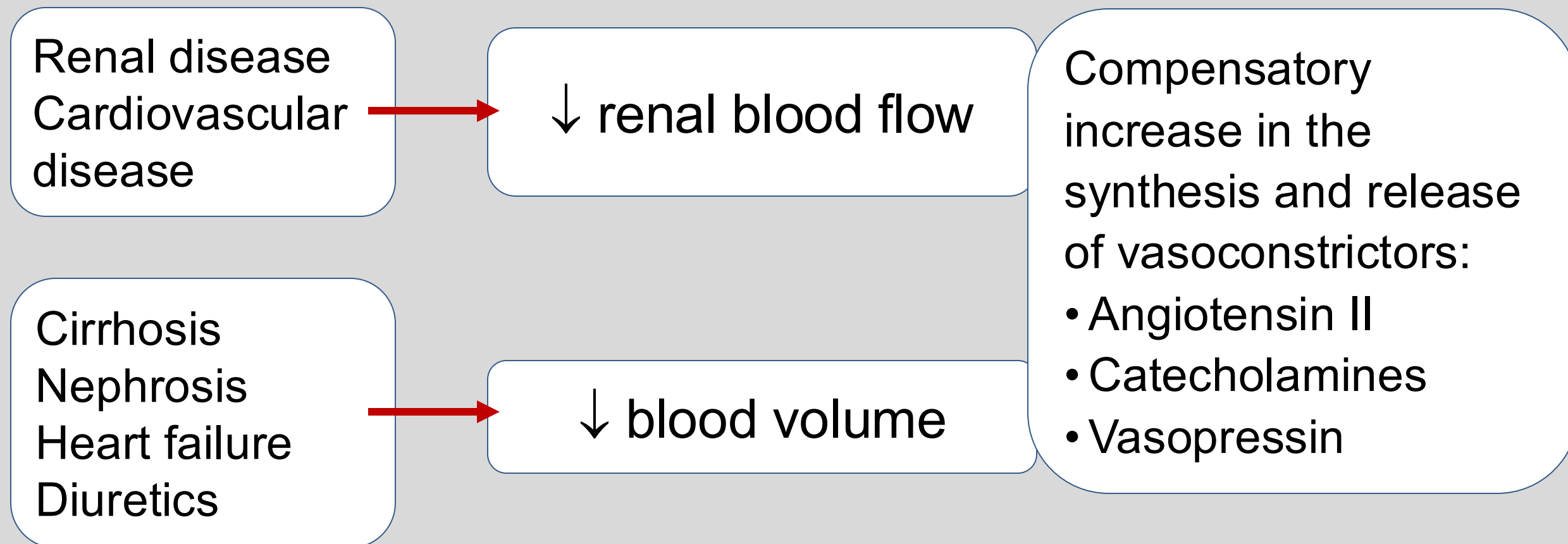
- hypovolemia
- congestive heart failure
- hepatic cirrhosis
- chronic kidney disease
- other states of activation of the sympatho-adrenal or renin-angiotensin system

The TAKEAWAY:

NSAID administration in patients with volume-contracted states and other states of activation of the sympathoadrenal or renin-angiotensin-aldosterone system (RAAS) removes the autoregulatory effects of prostanoids, resulting in a major ***reduction in RBF and GFR. Acute kidney injury*** can result.

Prostaglandins normally counteract intrarenal effects of vasoconstrictors.

Disease states leading to reduced renal blood flow or intrarenal vasoconstriction



Other effects of prostaglandin inhibition

- **CNS:**

headache; dizziness; tinnitus

- **Dermatologic:**

rash; pruritis; ecchymosis; purpura

- **Hematologic:** (rare)

agranulocytosis, thrombocytopenia, aplastic anemia

Practice point

A thorough patient evaluation to identify risk factors for toxicities

- GI,
- CV,
- acute kidney injury,
- and other toxicities

can minimize NSAID adverse drug events for the individual patient.

Why focus on NSAIDs' adverse effects?

FYI: Approximately 11% of hospital admissions for adverse drug events are for nonaspirin NSAID GI, renal, neurological, and allergic adverse effects.

(Howard RL, Avery AJ, Slavenburg S, et.al. Which drugs cause preventable admissions to hospital? A systematic review. Br J Clin Pharmacol. 2007 Feb.;(63)2:136-47. Epub 2006 Jun 26)

Check your knowledge

How does NSAID use increase the risk of bleeding?

What organ system is most frequently injured by NSAID use?

Check your knowledge

Which NSAIDs subclass has a dose-dependent lower risk of GI adverse effects?

Chronic or repeated NSAID use can lead to acute kidney injury by what mechanisms?

What are the various mechanisms of cardiovascular adverse effects?

- NSAIDs in Pregnancy and Lactation
- NSAIDs-related Drug Interactions and Contraindications
–drug-disease and drug-drug–

AVOID at beginning at 20 weeks gestation

If NSAID is necessary between 20-30 weeks (benefits to the mother outweigh the risk to fetus), use the lowest effective dose for shortest duration of time.

NSAID use may cause fetal renal dysfunction leading to:

- oligohydramnios and, in some cases,
- neonatal renal impairment, which may be fatal.

Oligohydramnios=too little amniotic fluid
Oligohydramnios can cause several structural and developmental adverse effects, including pulmonary hypoplasia, limb contractures, and facial compression deformities.

These effects are usually observed after prolonged (days to weeks) NSAID use but may occur as soon as 48 hours after NSAID initiation. If it occurs, oligohydramnios typically resolves within several days upon NSAID discontinuation.

AVOID after 30 weeks gestation

Risk of premature closure of the fetal ductus arteriosus → pulmonary hypertension, right-sided heart failure, ascites, pleural effusion (fetal hydrops)

These recommendations may not apply to the use of low-dose aspirin (81 mg) when used under the direction of a health care professional.

1st Trimester	Not strongly associated with birth defects (conflicting data)
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NSAIDs Class Drug-disease interactions

General Precautions

- Elderly
- Patients with cardiovascular risk factors
 - Hypertension
 - Fluid retention
 - Heart failure
- Coagulation disorder
- GI ulcer / bleeding
- Severe hepatic or renal impairment

CONTRAINDICATIONS

- Asthma (see aspirin)
- Coronary artery bypass graft
 - Before, during, and immediately after
- Bariatric surgery (nsNSAIDs)
 - Reduced mucosal defenses from the surgery is magnified with NSAID use
- Pregnancy
- Hypersensitivity reaction to aspirin or any other NSAID

NSAIDs Class Drug-Drug Interactions

Mechanisms on the following slides

- RAS Inhibitors
 - Angiotensin converting enzyme inhibitors (ACE inhibitors)
 - Angiotensin receptor blockers (ARBs)
 - Direct renin inhibitors (DRIs)
- Diuretics
- Antihypertensive drugs

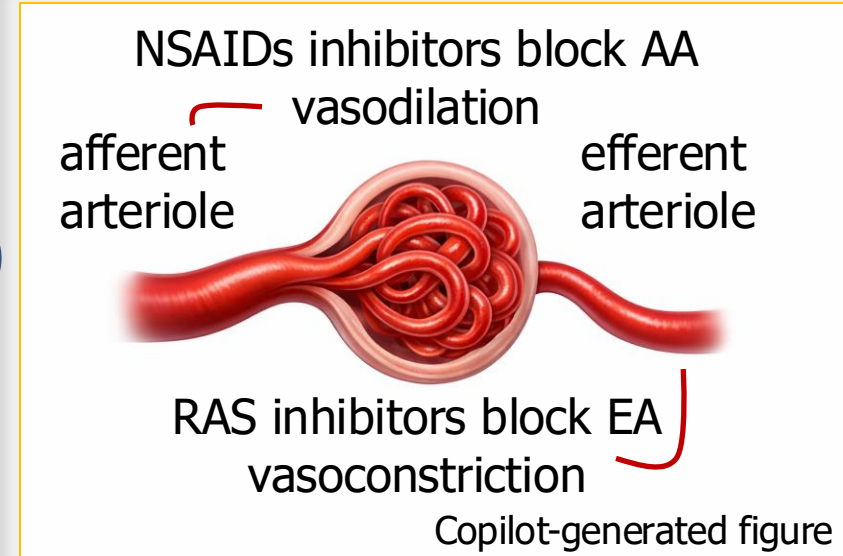
- Anticoagulants
- Antiplatelet drugs
- Drugs that cause mucosal injury
- Selective serotonin reuptake inhibitors (SSRIs)
- Lithium

Renin-Angiotensin System (RAS) inhibitors:

Angiotensin converting enzyme inhibitors (ACE inhibitors)

Angiotensin receptor blockers (ARBs)

Direct renin inhibitors (DRIs)



1. RAS inhibitors block ANG II's vasoconstrictor effects on the efferent arteriole



decrease glomerular capillary pressure



may cause acute kidney injury
In vulnerable patient

1. RAS inhibitors cause K⁺ reabsorption
2. NSAIDs increase K⁺ reabsorption



hyperkalemia → arrhythmia

NSAIDs cause Na⁺ and water reabsorption
↓
Increase intravascular volume



Blunt antihypertensive effects of the RAS inhibitors

Mechanisms of NSAIDs Drug-Drug Interactions

Diuretics
for treatment of
hypertension and
edema

- Diuretics increase Na^+ and water excretion. NSAIDs may attenuate this effect.
- Some diuretics stimulate COX activity. NSAIDs block the synthesis of vasodilating PGs, thus reduce diuretic efficacy.
- Some diuretics cause potassium reabsorption (potassium sparing diuretics) → risk of hyperkalemia with NSAIDs

Antihypertensive
drugs

NSAIDs increase Na^+ and water reabsorption, which can blunt the blood pressure lowering effects of the antihypertensive agents.

Anticoagulants,
Antiplatelet drugs,
Fibrinolytics

Increase the risk of bleeding (brain, GI tract, anywhere)

Mechanisms of NSAIDs Drug-Drug Interactions

Drugs that cause injury to the intestinal mucosa	↑ risk of gastrointestinal adverse events (alcohol; glucocorticoids; aspirin; others)
Selective serotonin reuptake inhibitors SSRIs are antidepressants	• Serotonin is taken up by platelets via serotonin transporters and stored in the platelets. • Serotonin is released from platelets (along with TXA2) and stimulates platelet aggregation. • SSRIs inhibit uptake of serotonin into platelets → ↓ amount of serotonin released from activated platelets → ↓ platelet aggregation → ↑ risk of bleeding • NSAIDs also ↑ risk of bleeding by inhibiting TXA2 synthesis.
Lithium (Li ⁺) (for bipolar disorder)	NSAIDs may increase Li ⁺ levels by increasing Li ⁺ reabsorption. Lithium has a narrow therapeutic window and significant toxicity.

Check your knowledge

How does maternal chronic NSAID use affect the fetus?

Sum up the mechanisms of NSAIDs-related drug interactions in two sentences.

Summary

Arachidonic Acid → COX → Prostanoids Pathway

- Mechanical, inflammatory, or hormonal signals activate PLA2 → arachidonic acid
- COX-1 and COX-2 convert AA to PGG2 → PGH2 → prostanoids by specific prostaglandin synthases.
- PGE2: fever, pain sensitization, gastric protection, renal blood flow
- PGI2 = prostacyclin: vasodilation, inhibition of platelet aggregation
- TXA2 = thromboxane A2: (platelet activation, vasoconstriction)
- PGF2α: smooth muscle contraction
- PGD2 : allergic responses, sleep regulation

COX-1 vs COX-2: Functional Distinctions

- COX-1 produces prostanoids for constitutive actions: GI cytoprotection, platelet aggregation, and renal perfusion autoregulation
- COX-2 produces prostanoids for induceable actions: Inflammation, pain, fever and constitutive actions in brain, kidney, and endothelium

NSAIDs Class Effects

PK: NSAIDs are chemically diverse weak acids.

- A: Well absorbed
- D: Highly bound to plasma proteins and widely distributed in tissues
- M: Mainly via CYP3A or 2C enzymes with biliary excretion and enterohepatic circulation. E: Metabolites active drug are excreted in urine
- $t_{1/2}$: Half-lives vary from <6 hours) to >10 hours.

Therapeutic effects:

- Analgesic,
- Anti-inflammatory,
- Antipyretic

Efficacy:

- All NSAIDs have similar effectiveness in the management of mild to moderate pain, fever, and inflammation. Some patients seem to respond better to one drug than to another.

NSAIDs Class Adverse Effects

- GI: loss of cytoprotection → dyspepsia, ulcers, bleeding, perforation
 - Risk is increased by age, glucocorticoids, anticoagulants, SSRIs, and prior ulcer
- CV:
 - Inhibition of TXA2-mediated platelet aggregation → reduced formation of platelet plug → increased risk of bleeding
 - Inhibiting PGI2-mediated platelet disaggregation allows unopposed TXA2-mediated platelet aggregation → prothrombotic effect → increased risk myocardial infarction, stroke
 - nsNSAIDs: COX2 inhibition is more sensitive than TXA2 inhibition → imbalance towards thrombus formation
 - COX-2 selective agents inhibit endothelial COX2 but not TXA2 → imbalance towards thrombus formation
- Renal: Inhibition of COX-1 and COX-2 derived prostanoids → ↓ renal perfusion and GFR, Na⁺ and water reabsorption (↑ blood pressure / edema → ↑ risk of MI, stroke / heart failure), and hyperkalemia

NSAIDs Class Adverse Effects

- Bronchospasm related to aspirin- or NSAID-exacerbated respiratory disease may occur in patients with concomitant chronic rhinosinusitis and nasal polyposis. These patients should avoid all NSAIDs.
- CNS effects include headache, dizziness, and reversible tinnitus.
- Rashes and urticaria can occur but severe dermal reactions are rare.
- Elevated serum transaminases are common, but liver failure is rare.
- Pregnancy:
 - Oligohydramnios (from 20 weeks gestation) and
 - Closure of patent ductus arteriosus (third trimester)

Check your knowledge

What membrane-derived fatty acid is the precursor of the prostanoids?

- Arachidonic acid (AA)

Where in the cell is the precursor located?

- AA is a component of membranes of the phospholipids that make up the nuclear-endoplasmic reticulum complex.

What is the process that converts the precursor into the five biologically active prostanoids?

- Cyclooxygenases (COX-1 and COX-2) convert AA to PGH₂ in a two-step process, which is then converted to the biologically active prostanoids by prostanoid-specific synthase enzymes.

Name the five prostanoids.

- Prostanoids: A subclass of the large family of eicosanoid signaling molecules derived from AA and other 20-carbon fatty acids
- Prostaglandins: PGE₂, PGF₂ α , PGD₂
- Other prostanoids: PGI₂ (prostacyclin), TXA₂ (thromboxane A₂)

Check your knowledge

What determines whether certain prostanoids are constitutively expressed or inducible? (and define the terms)

- Constitutive expression: An enzyme (or receptor) that remains in an active state (always on) without requiring a stimulus
- Inducible expression: An enzyme (or receptor) that is activated only in response to a specific stimulus
- COX-1 is constitutively active at basal levels. Prostanoids produced by COX-1 primarily provide homeostatic (housekeeping) functions. PGE2 and PGI2 are widely expressed.
- COX-2 is inducible in response to inflammatory stimuli, including $\text{TNF}\alpha$, IL-1, growth factors, bacterial endotoxin producing inflammation, pain sensitization, and fever response.
- REMEMBER: **Both COX-1 and COX-2** contribute to the generation of **autoregulatory** and **homeostatic** prostanoids, and both can contribute to prostanoid formation during **inflammation**.

How do the prostanoids exert their different effects in various body tissues?

- Prostanoids are autocrine and paracrine molecules that bind their specific GPCRs, activating intracellular signaling.
- Cells express specific prostanoid receptors according to their physiological functions.

Check your knowledge

How do NSAIDs inhibit the synthesis of biologically active prostanoids?

- NSAIDs target COX-1 and COX-2, the enzymes involved in conversion of arachidonic acid to prostanoid precursors, thereby preventing the synthesis of prostanoids involved in many homeostatic and inflammatory functions.

How does the NSAIDs' mechanism of action relate to their therapeutic and adverse effects?

- The analgesic, anti-inflammatory, and antipyretic effects are due to inhibiting COX-2-derived prostanoid synthesis.
- Adverse effects occur due to disruption of the protective homeostatic functions of prostanoids, leading to potentially serious adverse effects in at-risk individuals.

Why is it helpful to recognize the affinities of the individual NSAIDs for COX-1 and COX-2?

- Less severe GI adverse effects occur the more COX-2 selective NSAIDs because the GI cytoprotective actions are mediated by COX-1-derived prostanoids.

Check your knowledge

How does NSAID use increase the risk of bleeding?

- Inhibiting COX-1-derived TXA2 synthesis in platelets reduces platelet aggregation and local vasoconstriction, which reduces formation of the platelet plug.
- How does use of nsNSAIDs and COX-2-selective drugs increase the risk of myocardial infarction and stroke?
- COX-2 selective agents inhibit endothelial COX-2-derived PGI2 synthesis, which reduces platelet disaggregation and local vasodilation, shifting the balance to TXA-2-mediated formation of thrombi, which can lead to myocardial infarction or stroke.
- nsNSAIDs inhibit synthesis of both platelet TXA2 and endothelial PGI2. COX-2 is more sensitive to NSAID inhibition, particularly under inflammatory/shear stress conditions → imbalance favoring TXA-2-mediated formation of thrombi, which can lead to myocardial infarction or stroke.

What organ system is most frequently injured by NSAID use?

- GI system: inhibition of COX-1-derived PGE2 and PGI2 inhibit mucus and bicarbonate secretion, blood flow, and inhibit the acid reducing effects of PGs.
 - Remember: NSAIDs reduce repair mechanisms and increase the risk of bleeding.

Check your knowledge

Which NSAIDs subclass has a dose-dependent lower risk of GI adverse effects?

- COX-2 selective agents do not inhibit COX-1-mediate cytoprotective effects.
- More COX-2 selective NSAIDs have lower affinity for COX-1-derived prostaglandins

Chronic or repeated NSAID use can lead to acute kidney injury by what mechanisms?

- Inhibiting the autoregulatory vasodilation of the afferent arteriole by PGE2 / PGI2 reduces renal perfusion and glomerular filtration rate.
- Patients with volume depletion, chronic kidney disease, heart failure, or cirrhosis are in states of sympatho-adrenal and renin-angiotensin-aldosterone upregulation are at highest risk for AKI.

What are the various mechanisms of cardiovascular adverse effects?

- Inhibition of platelet TXA2 synthesis increases the risk of thrombotic myocardial infarction and stroke.
- Inhibition of renal Na⁺ and water excretion → increased Na⁺ and water reabsorption → increased blood volume, peripheral edema, and hypertension → increases blood pressure → the prothrombotic risk together with fluid retention can lead to myocardial infarction and stroke
- Hyperkalemia increases the risk of arrhythmia.

Check your knowledge

How does maternal chronic NSAID use affect the fetus?

- Oligohydramnios: Amniotic fluid is formed from fetal urine. From 20 weeks to term, reduced fetal renal function → reduced urine production → oligohydramnios → adverse fetal effects
- Closure of PDA: COX-2-derived PGE₂ maintains the patent ductus arteriosus. NSAID use in the third trimester can potentially cause closure of the PDA → fetal pulmonary hypertension, right-sided heart failure, and fetal hydrops

Sum up the mechanisms of NSAIDs-related drug interactions in two sentences.

- Concurrent use of drugs that have the same effects (therapeutic or adverse effects) enhance NSAIDs-related adverse effects: GI irritants, drugs that increase bleeding (anticoagulants and antiplatelet agents), drugs that modify blood pressure and/or renal perfusion, drugs with CNS side effects
- Drugs that modify clearance of NSAIDs / NSAIDs that modify clearance of other drugs



Careful Prescribing

patient centered

safe, effective, and necessary

the lowest effective dose for the shortest duration possible

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